

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Fatty Acids - Washed RBCs (FAR_L)

Data File: FAR_L.xpt

First Published: December 2025

Last Revised: NA

Component Description

The analysis of red blood cell (RBC) fatty acids (FA) is an indicator of long term (4 months) FA status. Evaluation of RBC fatty acid content has become increasingly favored as a measure of n-3 polyunsaturated fatty acids (PUFA) intake. PUFA alter membrane physical characteristics and the activity of membrane-bound proteins. In membranes, they interact with ion channels and can be converted into bioactive eicosanoid (Harris, et al., 2008).

The American Heart Association recommendation for those with no history of coronary heart disease (CHD) is to eat two fatty fish servings per week and for those with a history of CHD to consume 1 g/day of two omega-3 fatty acids, eicosapentaenoic (EPA) and docosahexaenoic (DHA) acid (Kris-Etherton, et al., 2002). The Omega-3 Index is a measure of EPA (C20:5 n-3) and DHA (C22:6 n-3) in the blood, and expressed as a percent of 21 fatty acids, in specifically the red blood cell membranes (Harris, et al., 2004). The Omega-3 Index has been suggested for use to identify those at risk for death from coronary heart disease (Harris, et. al., 2004).

Eligible Sample

Examined participants aged 6 years and older were eligible.

Description of Laboratory Methodology

Esterified FA are hydrolyzed from washed RBC membranes using sequential treatment with mineral acid and base in the presence of heat. Using a modification of Lagerstedt et al., total FA are hexane-extracted from the matrix (100 µL 50:50 saline diluted RBC) along with an internal standard solution containing thirteen stable isotopically-labeled FA to account for recovery (Lagerstedt et. al., 2001). The extract is derivatized with pentafluorobenzyl bromide (PFBBR) in the presence of triethylamine to form pentafluorobenzyl esters. The reaction mixture is injected onto a capillary gas chromatograph column to resolve individual FA of interest from other matrix constituents. FA are detected using electron capture negative-ion mass spectrometry within 34 minutes. Six saturated FA (SFA), four monounsaturated FA (MUFA), and eleven polyunsaturated FA (PUFA) (21 FA in total) are measured using selected ion monitoring. Quantitation is accomplished by comparing the peak area ratio of the analyte to internal standard in the unknown sample with the peak area ratio of a known amount of analyte to internal standard in a calibrator solution.

Refer to the Laboratory Method Files section for a detailed description on the laboratory methods used.

Laboratory Method Files

[Fatty Acids – RBC](#) (December 2025)

Laboratory Quality Assurance and Monitoring

Washed red blood cell specimens are processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to the National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured QA evaluation during unscheduled visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all the contract laboratories, which outlined the use of Westgard rules (Westgard, et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences' QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data may require additional demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews.

The [Fasting Questionnaire File](#) includes auxiliary information, such as fasting status, length of fast, and the time of venipuncture.

This laboratory data file can be linked to the Demographics file and other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. The lower limit of detection (LLOD, in mg/L) for the RBC fatty acids are shown below.

Variable Name	Analyte Description	LLOD
LBXPAN	alpha-Linolenic acid (C18:3n3) (%)	0.03
LBXP1A	Arachidic acid (C20:0) (%)	0.30
LBXPRA	Arachidonic acid (C20:4n6) (%)	0.08
LBXPDA	Docosanoic acid (C22:0) (%)	0.10
LBXPHA	Docosahexaenoic acid (C22:6n3) (%)	0.20
LBXPD3	Docosapentaenoic acid 3 (C22:5n3) (%)	0.07
LBXPD6	Docosapentaenoic acid 6 (C22:5n6) (%)	0.08
LBXPTA	Docosatetraenoic acid (C22:4n6) (%)	0.05
LBXPED	11,14-Eicosadienoic acid (C20:2n6) (%)	0.05
LBXP1E	11-Eicosenoic acid (C20:1n9) (%)	0.06
LBXPPE	Eicosapentaenoic acid (C20:5n3) (%)	0.40
LBXPLG	gamma-Linolenic acid (C18:3n6) (%)	0.07
LBXPGH	homo-gamma-Linolenic acid (C20:3n6) (%)	0.06
LBXP1G	Tetracosanoic acid (C24:0) (%)	0.10
LBXPNL	Linoleic acid (C18:2n6) (%)	0.30
LBXPMR	Myristic acid (C14:0) (%)	1.00
LBXPNR	15-Tetracosenoic acid (C24:1n9) (%)	0.05
LBXPOL	Oleic acid (C18:1n9) (%)	0.30
LBXPPL	Palmitoleic acid (C16:1n7) (%)	0.05
LBXPPM	Palmitic acid (C16:0) (%)	3.00
LBXPST	Stearic acid (C18:0) (%)	9.00

This component does not report data as mass concentrations (mg/L). Instead, data are reported as weight percentages of total fatty acids (see section on Calculating Percent RBC Fatty Acid). Two variables are provided for each of these analytes. The variable name ending in "LC" (ex., LBDPANLC) indicates whether the mass concentration result used in the weight percentage calculation was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBXPAN) provides the analytic weight percentage result for that analyte. For analytes with mass concentration results below the lower limit of detection (ex., LBDPANLC=1), the weight percentage values were set to missing because the weight percentage data cannot be compared to the LLOD. 15 mass concentration results in the dataset were below the LLOD and corresponding weight percentage values were set to missing and their comment variables set to 1.

Calculating Percent RBC Fatty Acid

Fatty acid concentrations measured in mg/L are summed for the total FA concentration (mg/L). Individual fatty acid weight percentages are calculated by dividing the individual fatty acid concentrations (mg/L) by the total FA (mg/L). The fatty acid data is expressed as weight percentages of total fatty acids.

References

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- Harris WS and vonSchacky C. The Omega-3 Index: a new risk factor for death from coronary heart disease? *Prev Med* 2004; 39:212-20.

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- Kris-Etherton PM, Harris WS, Appel LJ; American Heart Association. *Circ* 2002; 106:2747-2757.
- Lagerstedt S, Hinrichs D, Batt S, Magera M, Rinaldo P, McConnell J. Quantitative Determination of Plasma C8-C26 Total Fatty Acids for the Biochemical Diagnosis of Nutritional and Metabolic Disorders. *Mol Genet & Metab* 2001; 73: 38-45.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 6 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR
SAS Label: Phlebotomy 2 Year Weight
English Text: Phlebotomy 2 Year Weight
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 241728.85724	Range of Values	8068	8068	
.	Missing	0	8068	

LBXPAN - alpha-Linolenic acid (C18:3n-3) (%)**Variable Name:** LBXPAN**SAS Label:** alpha-Linolenic acid (C18:3n-3) (%)**English Text:** alpha-Linolenic acid (C18:3n-3) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.038 to 0.611	Range of Values	7176	7176	
.	Missing	892	8068	

LBDPANLC - alpha-Linolenic acid (C18:3n-3) cmt code

Variable Name: LBDPANLC
SAS Label: alpha-Linolenic acid (C18:3n-3) cmt code
English Text: alpha-Linolenic acid (C18:3n-3) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7176	7176	
1	below detectable limit	0	7176	
.	Missing	892	8068	

LBXP1A - Arachidic acid (C20:0) (%)**Variable Name:** LBXP1A**SAS Label:** Arachidic acid (C20:0) (%)**English Text:** Arachidic acid (C20:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.157 to 0.817	Range of Values	7133	7133	
.	Missing	935	8068	

LBDP1ALC - Arachidic acid (C20:0) comment code

Variable Name: LBDP1ALC
SAS Label: Arachidic acid (C20:0) comment code
English Text: Arachidic acid (C20:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7133	7133	
1	below detectable limit	0	7133	
.	Missing	935	8068	

LBXPRA - Arachidonic acid (C20:4n-6) (%)**Variable Name:** LBXPRA**SAS Label:** Arachidonic acid (C20:4n-6) (%)**English Text:** Arachidonic acid (C20:4n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4.17 to 21.2	Range of Values	7213	7213	
.	Missing	855	8068	

LBDPRALC - Arachidonic acid (C20:4n-6) comment code

Variable Name: LBDPRALC
SAS Label: Arachidonic acid (C20:4n-6) comment code
English Text: Arachidonic acid (C20:4n-6) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7213	7213	
1	below detectable limit	0	7213	
.	Missing	855	8068	

LBXPDA - Docosanoic acid (C22:0) (%)**Variable Name:** LBXPDA**SAS Label:** Docosanoic acid (C22:0) (%)**English Text:** Docosanoic acid (C22:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.629 to 3.26	Range of Values	7165	7165	
.	Missing	903	8068	

LBDPDALC - Docosanoic acid (C22:0) comment code

Variable Name: LBDPDALC
SAS Label: Docosanoic acid (C22:0) comment code
English Text: Docosanoic acid (C22:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7165	7165	
1	below detectable limit	0	7165	
.	Missing	903	8068	

LBXPHA - Docosahexaenoic acid (C22:6n-3) (%)**Variable Name:** LBXPHA**SAS Label:** Docosahexaenoic acid (C22:6n-3) (%)**English Text:** Docosahexaenoic acid (C22:6n-3) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.337 to 10.4	Range of Values	7215	7215	
.	Missing	853	8068	

LBDPHALC - Docosahexaenoic acid (C22:6n-3) cmt code

Variable Name: LBDPHALC

SAS Label: Docosahexaenoic acid (C22:6n-3) cmt code

English Text: Docosahexaenoic acid (C22:6n-3) comment code

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7215	7215	
1	below detectable limit	0	7215	
.	Missing	853	8068	

LBXPD3 - Docosapentaenoic acid 3 (C22:5n-3) (%)**Variable Name:** LBXPD3**SAS Label:** Docosapentaenoic acid 3 (C22:5n-3) (%)**English Text:** Docosapentaenoic acid 3 (C22:5n-3) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.381 to 7.5	Range of Values	7210	7210	
.	Missing	858	8068	

LBDPD3LC - Docosapentaenoic acid 3 (C22:5n-3) cmt**Variable Name:** LBDPD3LC**SAS Label:** Docosapentaenoic acid 3 (C22:5n-3) cmt**English Text:** Docosapentaenoic acid 3 (C22:5n-3) comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7210	7210	
1	below detectable limit	0	7210	
.	Missing	858	8068	

LBXPD6 - Docosapentaenoic acid (C22:5n-6) (%)**Variable Name:** LBXPD6**SAS Label:** Docosapentaenoic acid (C22:5n-6) (%)**English Text:** Docosapentaenoic acid (C22:5n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.116 to 1.66	Range of Values	7213	7213	
.	Missing	855	8068	

LBDPD6LC - Docosapentaenoic acid (C22:5n-6) cmt cd**Variable Name:** LBDPD6LC**SAS Label:** Docosapentaenoic acid (C22:5n-6) cmt cd**English Text:** Docosapentaenoic acid (C22:5n-6) comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7213	7213	
1	below detectable limit	0	7213	
.	Missing	855	8068	

LBXPTA - Docosatetraenoic acid (C22:4n-6) (%)**Variable Name:** LBXPTA**SAS Label:** Docosatetraenoic acid (C22:4n-6) (%)**English Text:** Docosatetraenoic acid (C22:4n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.563 to 8.89	Range of Values	7213	7213	
.	Missing	855	8068	

LBDPTALC - Docosatetraenoic acid (C22:4n-6) cmt cd

Variable Name: LBDPTALC
SAS Label: Docosatetraenoic acid (C22:4n-6) cmt cd
English Text: Docosatetraenoic acid (C22:4n-6) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7213	7213	
1	below detectable limit	0	7213	
.	Missing	855	8068	

LBXPED - 11,14-Eicosadienoic acid (C20:2n6) (%)

Variable Name: LBXPED**SAS Label:** 11,14-Eicosadienoic acid (C20:2n6) (%)**English Text:** 11,14-Eicosadienoic acid (C20:2n6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.123 to 0.628	Range of Values	7215	7215	
.	Missing	853	8068	

LBDPEDLC - 11,14-Eicosadienoic acid(C20:2n6) cmt cd**Variable Name:** LBDPEDLC**SAS Label:** 11,14-Eicosadienoic acid(C20:2n6) cmt cd**English Text:** 11,14-Eicosadienoic acid (C20:2n6) comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7215	7215	
1	below detectable limit	0	7215	
.	Missing	853	8068	

LBXP1E - 11-Eicosenoic acid (C20:1n-9) (%)**Variable Name:** LBXP1E**SAS Label:** 11-Eicosenoic acid (C20:1n-9) (%)**English Text:** 11-Eicosenoic acid (C20:1n-9) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.109 to 0.764	Range of Values	7035	7035	
.	Missing	1033	8068	

LBDP1ELC - 11-Eicosenoic acid (C20:1n-9) cmt code

Variable Name: LBDP1ELC
SAS Label: 11-Eicosenoic acid (C20:1n-9) cmt code
English Text: 11-Eicosenoic acid (C20:1n-9) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7035	7035	
1	below detectable limit	0	7035	
.	Missing	1033	8068	

LBXPPE - Eicosapentaenoic acid (C20:5n-3) (%)**Variable Name:** LBXPPE**SAS Label:** Eicosapentaenoic acid (C20:5n-3) (%)**English Text:** Eicosapentaenoic acid (C20:5n-3) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.063 to 6.92	Range of Values	7213	7213	
.	Missing	855	8068	

LBDPPELC - Eicosapentaenoic acid (C20:5n-3) cmt cd

Variable Name: LBDPPELC
SAS Label: Eicosapentaenoic acid (C20:5n-3) cmt cd
English Text: Eicosapentaenoic acid (C20:5n-3) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7213	7213	
1	below detectable limit	2	7215	
.	Missing	853	8068	

LBXPLG - gamma-Linolenic acid (C18:3n-6) (%)**Variable Name:** LBXPLG**SAS Label:** gamma-Linolenic acid (C18:3n-6) (%)**English Text:** gamma-Linolenic acid (C18:3n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.016 to 0.168	Range of Values	7176	7176	
.	Missing	892	8068	

LBDPLGLC - gamma-Linolenic acid (C18:3n-6) cmt code

Variable Name: LBDPLGLC
SAS Label: gamma-Linolenic acid (C18:3n-6) cmt code
English Text: gamma-Linolenic acid (C18:3n6) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7176	7176	
1	below detectable limit	0	7176	
.	Missing	892	8068	

LBXPGH - homo-gamma-Linolenic acid(C20:3n-6) (%)**Variable Name:** LBXPGH**SAS Label:** homo-gamma-Linolenic acid(C20:3n-6) (%)**English Text:** homo-gamma-Linolenic acid(C20:3n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.535 to 4.6	Range of Values	7215	7215	
.	Missing	853	8068	

LBDPGHLC - homo-gamma-Linolenic acid (C20:3n-6) cmt**Variable Name:** LBDPGHLC**SAS Label:** homo-gamma-Linolenic acid (C20:3n-6) cmt**English Text:** homo-gamma-Linolenic acid (C20:3n-6) comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7215	7215	
1	below detectable limit	0	7215	
.	Missing	853	8068	

LBXP1G - Tetracosanoic acid (C24:0) (%)**Variable Name:** LBXP1G**SAS Label:** Tetracosanoic acid (C24:0) (%)**English Text:** Tetracosanoic acid (C24:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3.05 to 9.48	Range of Values	7165	7165	
.	Missing	903	8068	

LBDP1GLC - Tetracosanoic acid (C24:0) comment code

Variable Name: LBDP1GLC
SAS Label: Tetracosanoic acid (C24:0) comment code
English Text: Tetracosanoic acid (C24:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7165	7165	
1	below detectable limit	0	7165	
.	Missing	903	8068	

LBXPNL - Linoleic acid (C18:2n-6) (%)**Variable Name:** LBXPNL**SAS Label:** Linoleic acid (C18:2n-6) (%)**English Text:** Linoleic acid (C18:2n-6) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4.04 to 15.6	Range of Values	7173	7173	
.	Missing	895	8068	

LBDPNLLC - Linoleic acid (C18:2n-6) comment code

Variable Name: LBDPNLLC
SAS Label: Linoleic acid (C18:2n-6) comment code
English Text: Linoleic acid (C18:2n-6) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7173	7173	
1	below detectable limit	0	7173	
.	Missing	895	8068	

LBXPMR - Myristic acid (C14:0) (%)**Variable Name:** LBXPMR**SAS Label:** Myristic acid (C14:0) (%)**English Text:** Myristic acid (C14:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.135 to 1.13	Range of Values	7153	7153	
.	Missing	915	8068	

LBDPMRLC - Myristic acid (C14:0) comment code

Variable Name: LBDPMRLC
SAS Label: Myristic acid (C14:0) comment code
English Text: Myristic acid (C14:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7153	7153	
1	below detectable limit	13	7166	
.	Missing	902	8068	

LBXPNR - 15-Tetracosenoic acid (C24:1n9) (%)**Variable Name:** LBXPNR**SAS Label:** 15-Tetracosenoic acid (C24:1n9) (%)**English Text:** 15-Tetracosenoic acid (C24:1n9) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.84 to 10.6	Range of Values	7166	7166	
.	Missing	902	8068	

LBDPNRLC - 15-Tetracosenoic acid(C24:1n9) comt code

Variable Name: LBDPNRLC
SAS Label: 15-Tetracosenoic acid(C24:1n9) comt code
English Text: 15-Tetracosenoic acid (C24:1n9) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7166	7166	
1	below detectable limit	0	7166	
.	Missing	902	8068	

LBXPOL - Oleic acid (C18:1n-9) (%)**Variable Name:** LBXPOL**SAS Label:** Oleic acid (C18:1n-9) (%)**English Text:** Oleic acid (C18:1n-9) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
7.93 to 19.1	Range of Values	7125	7125	
.	Missing	943	8068	

LBDPOLL - Oleic acid (C18:1n-9) comment code

Variable Name: LBDPOLL

SAS Label: Oleic acid (C18:1n-9) comment code

English Text: Oleic acid (C18:1n-9) comment code

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7125	7125	
1	below detectable limit	0	7125	
.	Missing	943	8068	

LBXPPL - Palmitoleic acid (16:1n-7) (%)**Variable Name:** LBXPPL**SAS Label:** Palmitoleic acid (16:1n-7) (%)**English Text:** Palmitoleic acid (16:1n-7) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.025 to 2.17	Range of Values	7084	7084	
.	Missing	984	8068	

LBDPPLLC - Palmitoleic acid (16:1n-7) comment code

Variable Name: LBDPPLLC
SAS Label: Palmitoleic acid (16:1n-7) comment code
English Text: Palmitoleic acid (16:1n-7) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7084	7084	
1	below detectable limit	0	7084	
.	Missing	984	8068	

LBXPPM - Palmitic acid (C16:0) (%)**Variable Name:** LBXPPM**SAS Label:** Palmitic acid (C16:0) (%)**English Text:** Palmitic acid (C16:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
15.4 to 31.1	Range of Values	7215	7215	
.	Missing	853	8068	

LBDPPMLC - Palmitic acid (C16:0) comment code

Variable Name: LBDPPMLC
SAS Label: Palmitic acid (C16:0) comment code
English Text: Palmitic acid (C16:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7215	7215	
1	below detectable limit	0	7215	
.	Missing	853	8068	

LBXPST - Stearic acid (C18:0) (%)**Variable Name:** LBXPST**SAS Label:** Stearic acid (C18:0) (%)**English Text:** Stearic acid (C18:0) (%)**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
10.1 to 21.8	Range of Values	7165	7165	
.	Missing	903	8068	

LBDPSTLC - Stearic acid (C18:0) comment code

Variable Name: LBDPSTLC
SAS Label: Stearic acid (C18:0) comment code
English Text: Stearic acid (C18:0) comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	detectable result	7165	7165	
1	below detectable limit	0	7165	
.	Missing	903	8068	